The Mentaculus and its Limits

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1 The Mentaculus and SMI

David Albert and Barry Loewer have proposed that all the laws and probabilities of the special sciences (e.g. biology, psychology, economics, etc.) can, in principle, be reduced to fundamental physics.[1] More specifically, they have proposed the following:

[1] Technically, to fundamental physics plus a set of statistical-mechanical postulates.

> **Statistical Mechanical Imperialism (SMI).** The laws and probabilities of the special sciences can be reduced to a set of postulates known as ***the Mentaculus***.[2]

[2] Albert (2012); Loewer (2020)

To see how the reduction works, let us look at an example.

1.1 An example

Take an arbitrary scientific "law". For instance, this law of biology:

> *With no predators and no scarcity of resources, a population of organisms grows exponentially.*

How is this recovered from physics? Like this:

1. Go back to the beginning of the universe. According to fundamental physics, the universe could have started in a great many different initial states. Take *all* of those states.
2. Keep those that evolve, by the laws of physics, into states in which there is a planet like the Earth, with organisms, and in which the population of a certain organism *O* has no predators and no limits on its resources.[3]
3. What the Mentaculus says is that **the vast majority** of these states are states in which the population of *O*s grows exponentially.

[3] Here *O* is a kind of organism. It could be deer, ducks, etc.

That is all. The biological law is recovered as a fact about *what proportion* of the possible initial states behave in a certain way.

1.2 *The postulates*

Each step of the example needs some postulate to be completed.

In step 1, I mentioned the *initial states* of the universe. These are fixed by **fundamental physics**. Since it concerns the interaction between microscopic particles, we will call it:

> **Microphysical Laws (ML).** The fundamental physical laws.

For simplicity, let us take ML to be the laws of N-particle Newtonian mechanics; little turns on this.[4]

In step 2, we keep the states in which there is a planet like the Earth, and there are organisms, etc. This in turn requires the universe to have **the macroscopic structure we are familiar with**. Albert and Loewer have shown that the following postulate is needed for that:

> **Past Hypothesis (PH).** The *initial macroscopic state* of the universe is one of extremely low entropy.[5]

In step 3, I say "the vast majority". For that to mean anything we need to be able to measure the size of a set of initial states. We need:

> **Statistical Postulate (PROB).** The *probability* that a given macroscopic state is realized by a given microscopic state is given by the uniform measure for that macroscopic state, conditional on PH.[6]

With PROB in place we can adopt the following definition:[7]

$$\text{The vast majority of states are states in which } P$$
$$\Updownarrow_{\text{def}}$$
$$\text{the conditional probability of } P \text{ is high.}$$

That equivalence is the whole content of step 3 of the example, and it is where the problem I care about lives: *"the vast majority" presupposes that we can measure*. So we have:

$$\textbf{Mentaculus} = \textbf{ML} + \textbf{PH} + \textbf{PROB}$$

[4] In Newtonian mechanics, the ***microphysical state*** (or microstate) of the universe is described by the position and velocity of each particle: it describes the universe *to the finest possible granularity*. Since Newtonian mechanics is *deterministic*, the initial state fixes the entire history, which is why in what follows "microstate" and "initial microstate" are interchangeable.

[5] A ***macroscopic state*** (or macrostate) is described by determining the locations, velocities and (macroscopic) attributes of the middle-sized and larger objects in the universe.

[6] In practice, and this is how I will use it: the uniform (Lebesgue) measure over the PH-compatible region, normalized (Loewer, 2001).

[7] For this definition to work we need, in addition, the following postulate:

Microphysical Supervenience (MPS): any two worlds that are physically possible according to ML and that agree on the microphysical state of the universe agree on all special-scientific facts.

This is what guarantees that the propositions of the *special sciences*—not only those of microphysics—can be represented as sets of initial microstates, which is necessary (but not sufficient) for PROB to assign them a probability.

1.3 Representational details

With the machinery in place, the probability of a proposition P according to the Mentaculus can be represented in a diagram:

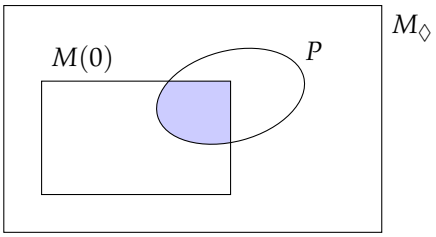


Figure 1: The Mentaculus' probability assignment.

Here:

- $M_\Diamond$ is the set of all possible initial microstates according to ML,
- $M(0)$ is the set of those in which PH is true, which is assigned a "size" of 1, and
- the proposition P is represented as the set of initial microstates that make it true.

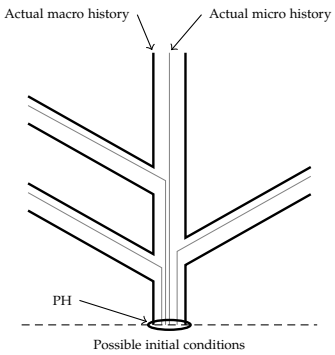


Figure 2: Under PH, micro histories evolve deterministically, while macro histories branch probabilistically (after Loewer, 2020).

2 *The Recipe*

What we did with the population example can be written down as a general procedure. Suppose some special science explains a fact E in terms of (a possibly very complex fact) F, and that we want to recover that explanation in terms of the Mentaculus. Then:

(S0) Represent E and F as sets of initial microstates. MPS guarantees that the representation is materially adequate.

(S1) Calculate the objective chance $\mathbf{Pr}(E|F)$ of E given F, using the measure $\mathbf{Pr}$ determined by PROB.

(S2) If this chance is high, we have recovered the explanation in terms of the Mentaculus. Done.

(S3) If not, we need to revise either the microphysical laws, or the special-scientific explanation, or both.

In the example of §1.1, F is "the population of Os has no predators and no limits on its resources" and E is "the population of Os grows exponentially"; step 3 is (S1), and we are assuming that (S2) holds and (S3) does not.

Two observations

1. This is *in-principle* explanation. No one claims that we can write down a physical formula characterizing the initial states F, or E.
2. The idea is that there are *physical features of the world* underwriting the explanations we give in biology, for instance by means of this law. Put differently, this is a law because it is *physically robust*.

3 *The problem: the PRR*

Step (S1) requires calculating

$$\mathbf{Pr}(E|F) = \frac{\mathbf{Pr}(E \cap F)}{\mathbf{Pr}(F)} = \frac{\lambda(E \cap F \cap M(0))}{\lambda(F \cap M(0))},$$

where λ is the measure PROB refers to: the uniform, or Lebesgue, measure.

The problem. According to standard measure theory, λ does not assign a value to every subset of $M(0)$: assuming the Axiom of Choice **(AC)**, there are subsets that are not Lebesgue measurable (Halmos, 1950, p. 67), and on those **Pr** is simply undefined. If $E \cap F$ or F is one of them, **the Recipe breaks down at step (S1)**: there is no probability to calculate.

Since SMI aims to recover *all* adequate special-scientific explanations by means of the Recipe, it is committed to the following principle, which functions as a **bridge** between the Mentaculus and the special sciences:

> **Probabilistic Representability Requirement (PRR).** For any adequate explanation of a fact E by means of the features F in some special science, the sets $E \cap F \cap M(0)$ and $F \cap M(0)$ are measurable by the probability measure determined by PROB.
>
> ***In plain words:*** *everything we say in the special sciences has to correspond, by some miracle, to a set of initial states to which the uniform measure (λ) assigns a size.*

And nothing in the standard formulation of the Mentaculus guarantees it: MPS guarantees that special-science propositions correspond to sets of microstates, not that those sets are *measurable*. **PH** is not probabilistic: it says nothing about measures. **PROB** already presupposes the Lebesgue measure in defining **Pr**; it cannot in addition guarantee its own domain of application.

The PRR is therefore an **additional and substantive** assumption, one that requires independent justification if SMI's explanatory ambitions are to be sustained.

3.1 *Why this is a new problem*

The Mentaculus and SMI have plenty of critics, but the criticisms usually focus on…

- whether the laws of the Mentaculus are as robust as those of the special sciences (Fodor, 1974),
- whether the predictions of the Mentaculus deserve the title of "laws" (Chen, 2023; Dorst, 2024),
- or whether those predictions are reliable outside the context of thermodynamics (Weslake, 2014).

My criticism is orthogonal to all of these questions: I am not asking whether the predictions of the Mentaculus are robust, deserve the title of "laws", or are reliable outside thermodynamics. *I am asking whether there are any predictions to judge at all.*

4 *Replies*

There are several ways one might try to secure the PRR. I will go through four: that we can *deny* AC, that non-measurable sets are *undefinable*, that they are *rare* or *heterogeneous*, and that they are *epistemically undecidable*. All of them fail. And two of them exhibit a pattern of circularity.

4.1 *Denying the Axiom of Choice*

Reply. Non-measurable sets exist only if the Axiom of Choice (AC) does. Reject AC, or suspend it here.

Problem.

- AC is standard, and accepted as fully legitimate in mathematical practice.[8]
- It is equivalent to independently intuitive claims.[9]
- Rejecting it just to secure SMI is *ad hoc*: SMI gives no independent reason to revise set theory.

[8] E.g. a proof of a Millennium prize problem using AC would be considered legitimate, other things being equal.

[9] E.g. the Cartesian product of a family of non-empty sets is non-empty.

4.2 *Non-measurable sets are undefinable*

Reply. In set theory, we can prove that non-measurable sets exist (with AC), but we cannot write down a sentence that expresses their membership conditions.[10] Analogously, we might hold that no natural-language sentence expresses a non-measurable set of initial conditions in the Mentaculus. Special-science sentences are part of natural language, so PRR secured.

Problem: the parallel does not hold.

- The reply uses the *set-theoretic* notion of definability to conclude something about our natural-language sentences. But

  definable in the Mentaculus $\neq$ expressible in *natural language*.

- Our practice tells against the identification: *we almost never define these sets*. Take the sentence from §1.1:

  > *The population of Os (deer, ducks, etc.) has no predators and no limits on its resources*

  Which set of initial microstates does it correspond to? No one can give a formula, in microphysical terms, whose extension is exactly that set, but the sentence is perfectly meaningful. In special sciences, this type of sentences are the rule, and not the exception.

[10] Precisely: if ZFC plus an inaccessible cardinal is consistent, ZFC cannot prove of any formula quantifying over reals, with real parameters, that it defines a non-measurable set (Solovay, 1970); and given large cardinals, every such set is measurable (Shelah and Woodin, 1990). In ZFC alone this can fail: if $V = L$, there is a definable non-measurable set.

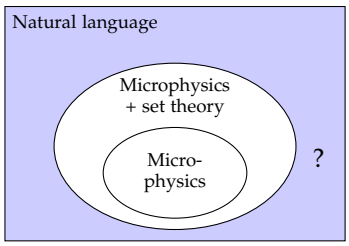


Figure 3: Microphysics $\subseteq$ microphysics + set theory (the Mentaculus) $\subseteq$ natural language. The reply needs the second inclusion to be an equality, so that the shaded region is empty. Nothing guarantees that.

4.3 "Non-measurable sets are rare"

Reply. Non-measurable sets are a technical oddity: exotic, scarce, and unlikely to be what a special-science proposition lands on.

Problem. They are not rare, according to the following two theorems (Proofs in the appendix.) The first refutes the claim that non-measurable sets are rare. The second blocks the most obvious way to patch the problem.

> **Theorem 1 (abundance).** If the Axiom of Choice (AC) is true, there are *as many non-measurable subsets as there are subsets in total.*
>
> **Theorem 2 (variety).** Identify the sets the Mentaculus cannot tell apart. Then there are at most $\mathfrak{c}$ measurable classes—as many as there are real numbers—and at least $2^{\mathfrak{c}}$ non-measurable ones—as many as there are sets of real numbers.

Moral. I need not show that some special-science proposition is non-measurable; the defender of SMI must show that *all* of them are measurable.

4.4 Non-measurable sets are heterogeneous or unnatural

Reply. The special sciences traffic only in natural properties, and there is no reason to expect those to have non-measurable microphysical correlates.

Problem 1: naturalness has to be stipulated.

- SMI promises to deliver special-science reduction from minimal background assumptions[11] that even anti-reductionist physicalists accept (Fodor, 1974; Weslake, 2014). That is what makes the promise remarkable.
- In accepting naturalness, the proponent of SMI loses grounds against the anti-reductionist physicalist, and thus makes their theory less attractive.
- And **Loewer himself** set out to do without naturalness[12]

[11] Here, the "background assumptions" are really just one: MPS. ML, PH, and PROB are part of the theory itself, and thus not background assumptions.

[12] E.g. he developed a best-system account that "doesn't depend on Lewis's account of natural properties or on taking naturalness as primitive" (Loewer, 2007, 313). So it is only natural (no pun intended) for defenders of the Mentaculus to reject this option.

Problem 2: heterogeneity has already been conceded.

- Non-measurability requires a set to be heterogeneous *from the point of view of ML*.[13]
- **That is exactly the sense in which everyone concedes special-science predicates are heterogeneous**: Fodor (1974) uses it against reduction; Loewer (2009) **explicitly concedes** it, and denies that it blocks reduction. Once conceded, heterogeneity relative to ML cannot work as an exclusion criterion.

[13] *Example*: F in §1.1 (the initial microstates that evolve into states where the *O*s have no predators and no limits on resources) is almost certainly heterogeneous from the point of view of the microphysics, because ecosystems are chaotic.

4.5 *Epistemic decidability*

Reply (David Albert). Non-measurable sets are excluded because special-science sentences must be *epistemically decidable*. Schematically:

- **Premise 1.** Sentences of the special sciences must satisfy the following constraint:

  > **Observability Constraint (OC).** A sentence ϕ satisfies OC iff it is possible to determine, by observation and analysis of the data, whether ϕ is true or false.

- **Premise 2.** OC is violated by sentences that correspond to non-measurable sets of microstates.

So sentences of special sciences must correspond to measurable sets of microstates. PRR secured.

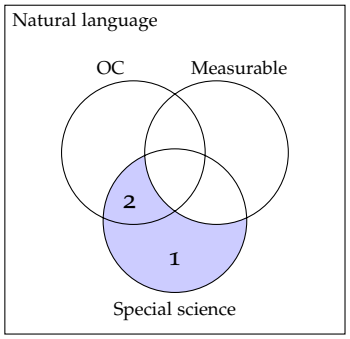


Figure 4: The reply needs the shaded region to be empty: no special-science sentence outside the measurable ones. Premise 1 empties region 1 (special science without OC); Premise 2 empties region 2 (OC without measurability). Problems 1 and 2 doubt each in turn.

Problem 1: Premise 1 is implausible on its own.

- It is implausible that epistemic decidability is a *necessary* condition for figuring in a special-scientific explanation.
- Our ordinary and scientific vocabularies are full of predicates whose application we cannot reliably determine in arbitrary cases; in fact, most of them are like that.

Problem 2: Accepting premise 2 begs the question.

- *Detection is itself a special-science affair*: what it is to competently apply a predicate, which procedures count as measurements, and when they are reliable are questions for linguistics, cognitive science, measurement theory, epistemology, biology, etc.
- So OC does its work only if *those* sciences—the ones concerned with verification—already reduce to the Mentaculus. **And that is precisely what was in question.**[14]

[14] A way out. Grant Premise 1, and take a special-science sentence ϕ that satisfies OC. For the reply to conclude that ϕ corresponds to a measurable set, the procedures that decide ϕ must themselves be representable in the Mentaculus. A Mentaculus-based theory of measurement, such as Albert's account of records and measurement in terms of PH (Albert, 2015, ch. 2, 36–39), would supply that. The reply then works, but at the price of reducing *one* special-science subject: measurement. That is an extra background assumption, though a less controversial one than naturalness; if the theory of measurement proves explanatory and elegant, the anti-reductionist might grant it.

4.6 The circularity pattern

Two of the replies share a single shape, and fail for the same reason. Each restricts the special-science sentences by appeal to something that is *itself* the subject matter of a special science:

Reply	**Appeals to…**	**But that is studied by…**
Undefinability (§4.2)	which sets natural-language sentences express	metasemantics, linguistics, etc.
Epistemic decidability (§4.5)	which sentences we can verify by observation and measurement	epistemology, cognitive science, measurement theory, etc.

In each case, for the reply to work one must assume that *that* stretch of the higher-level world can be represented within the microphysics in exactly the way the PRR asserts for the special sciences at large.

The Mentaculus cannot lean on the special sciences to secure its ability to encompass the special sciences.

5 Consequences

1. The Recipe stalls. Without the PRR, the Mentaculus assigns no probabilities to special-science propositions. Step (S1) cannot be carried out, and SMI's explanatory ambitions cannot be sustained. ***It is not that it gives the wrong answer: it gives no answer at all. This substantially limits the scope of the Mentaculus.***

2. The objection generalizes. Nothing in the argument turned on distinctive commitments of the Mentaculus. What did the work was a more general thesis:

Probability foundationalism. The probabilities of the special sciences derive from a probability measure supplied by physics; there are no autonomous special-science probabilities floating free of that measure.

The Mentaculus is the most developed and prominent implementation of this thesis, but ***any position of this kind must confront its own analogue of the PRR***, because any probability measure resembling it will have its own non-measurable sets.

3. Moral. Debates over the scope of unifying scientific frameworks must attend not only to where they succeed empirically, but to where they are able to issue predictions at all. And here the burden of proof lies on the other side: **it is up to the defender of SMI to show that the Mentaculus can issue verdicts outside thermodynamics.**

6 *Appendix: proofs*

Throughout this appendix I replace the normalized uniform measure on $M(0)$ with the Lebesgue measure on the unit interval $[0,1]$. This involves no relevant loss of generality: a positive-measure region of phase space, equipped with its normalized measure, is a standard atomless probability space, and every such space is measure-theoretically isomorphic, up to sets of measure zero, to $[0,1]$ equipped with the Lebesgue measure; the isomorphism carries measurable sets to measurable sets in both directions (Halmos 1950, §§41–43; Bogachev 2007, §9.2). So let λ be the Lebesgue measure on $[0,1]$.

Theorem 0 (Vitali). There are non-measurable subsets of $[0,1]$.

Proof. For $a, b \in [0,1]$, let $a \sim b$ iff $a - b \in \mathbb{Q}$. It is easily shown that $\sim$ is an equivalence relation, so each $a \in [0,1]$ belongs to an equivalence class $[a]_\sim$. Let $\mathcal{A}$ be the set of all equivalence classes. By the Axiom of Choice, there is a function $f : \mathcal{A} \to [0,1]$ such that $f([a]_\sim) \in [a]_\sim$ for all $a \in [0,1]$. Now define $V := \{f([a]_\sim) : a \in [0,1]\}$ and $V_q := \{x + q : x \in V\}$. We have two important facts:

Fact 1. The sets $\{V_q : q \in \mathbb{Q} \cap [-1,1]\}$ are pairwise disjoint.

Suppose $q_1 \neq q_2$ and $x \in V_{q_1} \cap V_{q_2}$. Then $x = v_1 + q_1 = v_2 + q_2$ for some $v_1, v_2 \in V$. But this would mean that $v_1 \sim v_2$, since $q_1, q_2 \in \mathbb{Q}$, contradicting the fact that V contains exactly one representative from each equivalence class.

Fact 2. $[0,1] \subseteq \bigcup\{V_q : q \in \mathbb{Q} \cap [-1,1]\} \subseteq [-1,2]$.

For the first inclusion, let $x \in [0,1]$. Then $x \in [x]_\sim$ and, by the construction of V, there is a unique $v \in V \cap [x]_\sim$. So $v \sim x$, which means $x = v + q$ for some $q \in \mathbb{Q}$, and hence $x \in V_q$. Since both x and v are in $[0,1]$, we have $|q| = |x - v| \leq 1$, and so $q \in \mathbb{Q} \cap [-1,1]$. Hence $x \in \bigcup\{V_q : q \in \mathbb{Q} \cap [-1,1]\}$. For the second inclusion, if $x \in V_q$ for some $q \in \mathbb{Q} \cap [-1,1]$, then $x = v + q$ for some $v \in V$; since $V \subseteq [0,1]$, this means $x = v + q \in [-1,2]$.

Now assume, for reductio, that V is measurable, with $\lambda(V) = \alpha$. Since λ is translation-invariant, $\lambda(V_q) = \alpha$ for every V_q. And since the V_q are pairwise disjoint, we have $\lambda(\bigcup\{V_q : q \in \mathbb{Q} \cap [-1,1]\}) = \sum_{q \in \mathbb{Q} \cap [-1,1]} \lambda(V_q) = \sum_{q \in \mathbb{Q} \cap [-1,1]} \alpha$. But this is α summed countably many times, which is either 0 or ∞. That is impossible, since Fact 2 gives us $1 \leq \lambda(\bigcup\{V_q : q \in \mathbb{Q} \cap [-1,1]\}) \leq 3$. ∎

Theorem 1 (abundance). If the Axiom of Choice is true, there are $2^{\mathfrak{c}}$ subsets of $[0,1]$ that are not λ-measurable.

Proof. We already have existence. Now let V be a non-measurable subset of $[0,1]$ and let C be the Cantor set. This set has cardinality $\mathfrak{c}$ and λ-measure 0. Since its cardinality is $\mathfrak{c}$, there are $2^{\mathfrak{c}}$ subsets S of C and, since λ is a complete measure, each of those subsets has λ-measure 0. Now consider the mapping

$$f : \mathcal{P}(C) \to \mathcal{P}([0,1]) \mid f(S) = V \triangle S,$$

where the symmetric difference $V \triangle S$ is defined as

$$V \triangle S := (V \setminus S) \cup (S \setminus V) = \{x \in [0,1] : x \in V \text{ or } x \in S, \text{ but not both}\}.$$

It is easily shown that f is injective, which means that there are at least $2^{\mathfrak{c}}$ sets of the form $V \triangle S$, with

$S \subseteq C$. Now, since S is measurable and measurability is closed under symmetric difference, $V \triangle S$ must be non-λ-measurable, since otherwise $V = (V \triangle S) \triangle S$ would be measurable. So the set $\{V \triangle S : S \subseteq C\}$ contains only non-λ-measurable sets and has cardinality $2^{\mathfrak{c}}$. ∎

Theorem 2 (variety). Define $A, B \subseteq [0,1]$ to be *λ-equivalent* (written $A \sim_\lambda B$) iff $A \triangle B$ is λ-measurable and $\lambda(A \triangle B) = 0$. For each $A \subseteq [0,1]$, let $[A]_\lambda$ be its λ-equivalence class, and define $[0,1]_\lambda = \{[A]_\lambda : A \subseteq [0,1]\}$. Then each equivalence class $[A]_\lambda$ contains either only λ-measurable sets or only non-λ-measurable sets. Moreover, assuming the Axiom of Choice, there are at most $\mathfrak{c}$ measurable equivalence classes and at least $2^{\mathfrak{c}}$ non-measurable ones.

Proof. First, we show that no equivalence class mixes measurable and non-measurable sets. Let $A, B \subseteq [0,1]$ be such that $A \sim_\lambda B$. Then $A \triangle B$ is λ-measurable and $\lambda(A \triangle B) = 0$. If A is λ-measurable, then $B = (A \triangle B) \triangle A$ is λ-measurable, since measurability is closed under symmetric difference. For the same reason, if A is not λ-measurable, then B cannot be λ-measurable, since otherwise $A = (A \triangle B) \triangle B$ would be λ-measurable, a contradiction. Hence every equivalence class $[A]_\lambda$ contains either only measurable or only non-measurable sets.

Next, we bound the number of measurable equivalence classes. By definition, the Lebesgue measure is the completion of the Borel measure, so every λ-measurable set differs from some Borel set by a measurable set of λ-measure zero. Therefore every measurable equivalence class contains a Borel representative. Since the family of Borel subsets of $[0,1]$ has cardinality $\mathfrak{c}$, there are at most $\mathfrak{c}$ measurable equivalence classes.

Finally, we show that there are at least $2^{\mathfrak{c}}$ non-measurable equivalence classes. Let $\mathcal{Z}$ be the family of null Borel subsets of $[0,1]$. Since there are only $\mathfrak{c}$ Borel sets, we have $|\mathcal{Z}| = \mathfrak{c}$ and hence $|\mathcal{Z} \times \mathfrak{c}| = \mathfrak{c}$. Using the well-ordering theorem, fix an enumeration $\{(N_\alpha, \kappa_\alpha) : \alpha < \mathfrak{c}\}$ of $\mathcal{Z} \times \mathfrak{c}$. Now, for each $\alpha < \mathfrak{c}$ we have $|[0,1] \setminus N_\alpha| = \mathfrak{c}$, so by transfinite recursion we can recursively choose a distinct point $x_\alpha \in [0,1] \setminus N_\alpha$ for each $\alpha < \mathfrak{c}$. We can then define a function $f : [0,1] \to \mathfrak{c}$ by setting $f(x_\alpha) = \kappa_\alpha$ and mapping all remaining points of $[0,1]$ to (say) 0. For each $\kappa < \mathfrak{c}$, let $F_\kappa := f^{-1}(\{\kappa\})$. Now, for every null Borel set $N \in \mathcal{Z}$ and every $\kappa < \mathfrak{c}$, we have $F_\kappa \nsubseteq N$. Indeed, since $(N, \kappa) \in \mathcal{Z} \times \mathfrak{c}$, there is some α with $(N_\alpha, \kappa_\alpha) = (N, \kappa)$, which means that $x_\alpha \in F_\kappa$ and $x_\alpha \notin N$. Now, for each $I \subseteq \mathfrak{c}$, define $A_I = \bigcup_{\kappa \in I} F_\kappa$. If $I \neq J$, there is some $\kappa' \in I \triangle J$. Then $F_{\kappa'} \subseteq A_I \triangle A_J$. If $A_I \triangle A_J$ were contained in some null Borel set N, then $F_{\kappa'} \subseteq N$, contradicting the earlier result. Therefore $A_I \triangle A_J$ is not contained in any null Borel set. But since λ is regular and complete, every measurable set of λ-measure zero is contained in a null Borel set; hence $A_I \triangle A_J$ is not λ-null, which means that A_I and A_J are not λ-equivalent. Thus the map $I \mapsto [A_I]_\lambda$ is injective from $\mathcal{P}(\mathfrak{c})$ into $[0,1]_\lambda$, and so $|[0,1]_\lambda| \geq 2^{\mathfrak{c}}$. To conclude, the family $\{[A_I]_\lambda : I \subseteq \mathfrak{c}\}$ has cardinality $2^{\mathfrak{c}}$ and, by the bound established above, at most $\mathfrak{c}$ of its members can be measurable equivalence classes. Since $\mathfrak{c} < 2^{\mathfrak{c}}$, removing at most $\mathfrak{c}$ classes from a family of cardinality $2^{\mathfrak{c}}$ leaves a family of cardinality $2^{\mathfrak{c}}$. ∎

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